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ATP107

P-Channel Power MOSFET -40V, -50A, 17mΩ, Single ATPAK

Features

- Low ON-resistance
- Slim package
- Halogen free compliance
- Large current
- 4.5V drive
- Protection diode in

Specifications

Absolute Maximum Ratings at Ta=25°C

Parameter	Symbol	Conditions	Ratings	Unit
Drain-to-Source Voltage	V _{DSS}		-40	V
Gate-to-Source Voltage	V _{GSS}		±20	V
Drain Current (DC)	I _D		-50	A
Drain Current (PW≤10μs)	I _{DP}	PW≤10μs, duty cycle≤1%	-150	A
Allowable Power Dissipation	P _D	Tc=25°C	50	W
Channel Temperature	T _{ch}		150	°C
Storage Temperature	T _{stg}		-55 to +150	°C
Avalanche Energy (Single Pulse) *1	E _{AS}		80	mJ
Avalanche Current *2	I _{AV}		-25	A

Note : *1 V_{DD}=-10V, L=200μH, I_{AV}=-25A

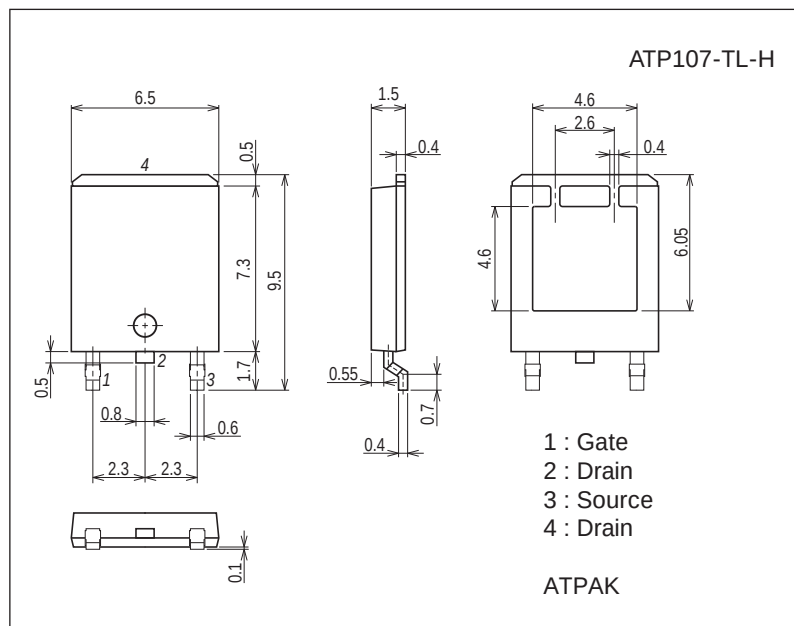
*2 L≤200μH, Single pulse

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

Package Dimensions

unit : mm (typ)

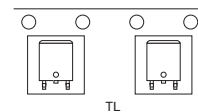
7057-001



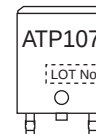
Product & Package Information

- Package : ATPAK
- JEITA, JEDEC : -
- Minimum Packing Quantity : 3,000 pcs./reel

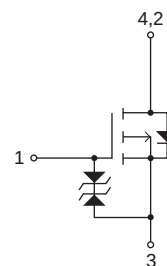
Packing Type: TL



Marking



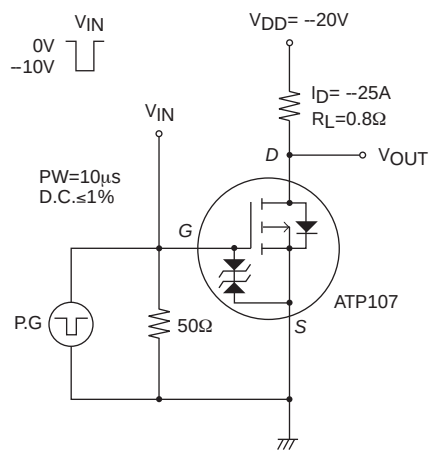
Electrical Connection



Electrical Characteristics at Ta=25°C

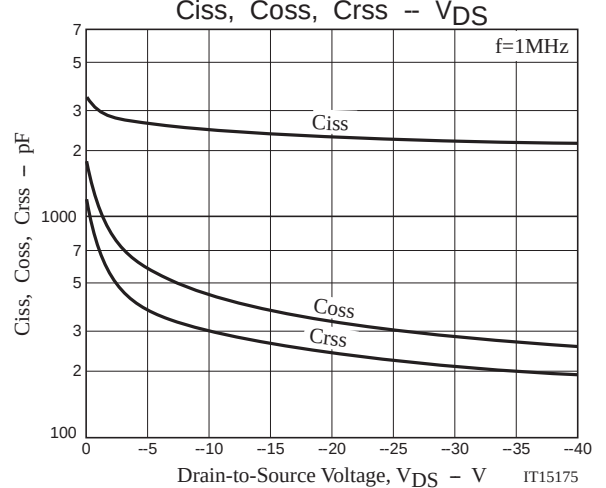
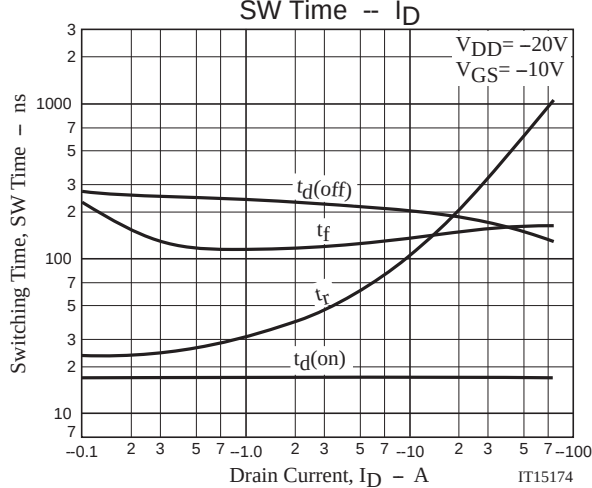
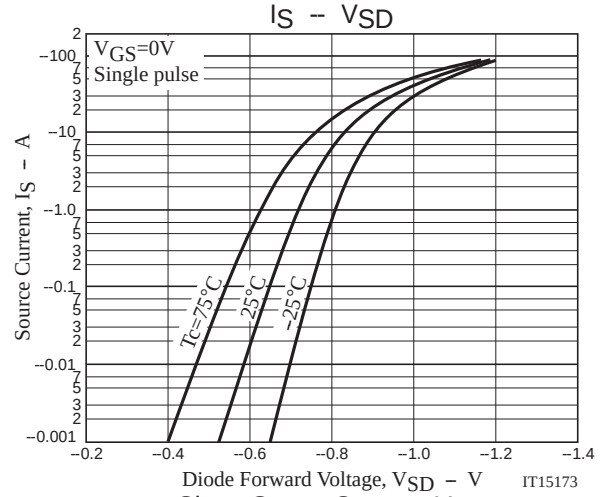
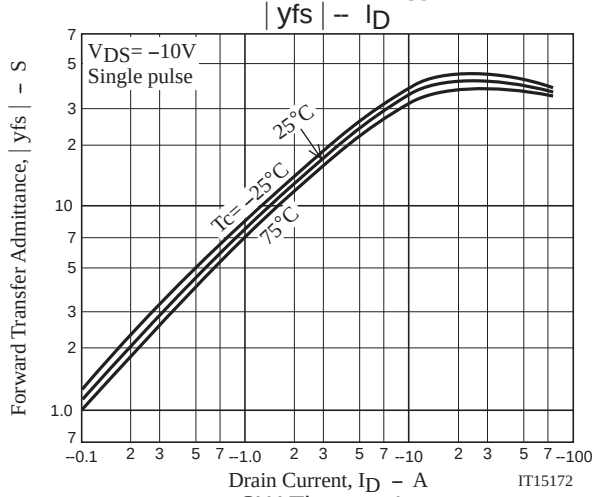
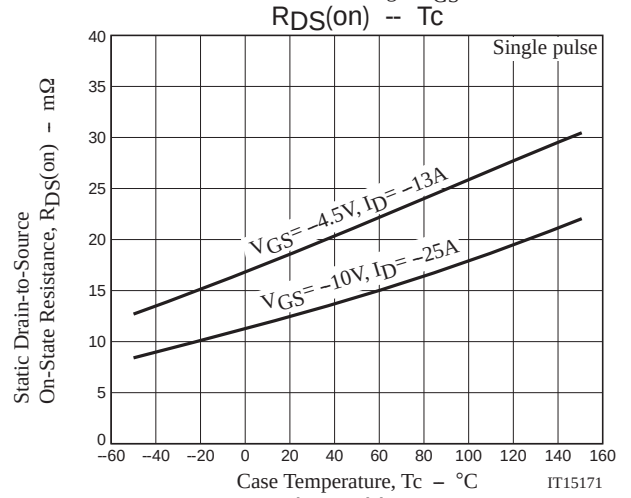
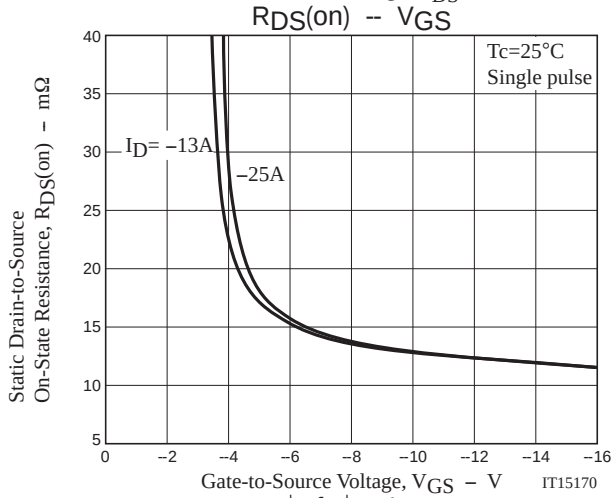
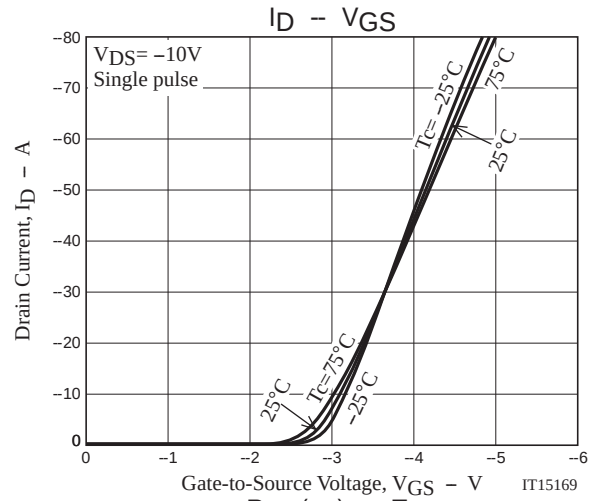
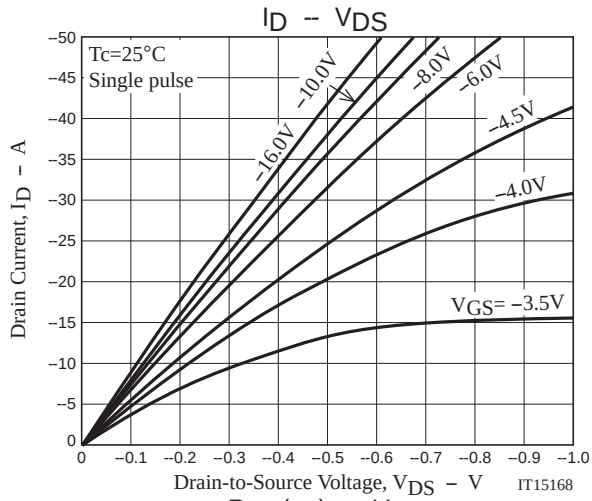
Parameter	Symbol	Conditions	Ratings			Unit
			min	typ	max	
Drain-to-Source Breakdown Voltage	$V_{(BR)DSS}$	$I_D=-1mA, V_{GS}=0V$	-40			V
Zero-Gate Voltage Drain Current	I_{DSS}	$V_{DS}=-40V, V_{GS}=0V$			-1	μA
Gate-to-Source Leakage Current	I_{GSS}	$V_{GS}=\pm 16V, V_{DS}=0V$			± 10	μA
Cutoff Voltage	$V_{GS(off)}$	$V_{DS}=-10V, I_D=-1mA$	-1.5		-2.6	V
Forward Transfer Admittance	$ y_{fs} $	$V_{DS}=-10V, I_D=-25A$		40		S
Static Drain-to-Source On-State Resistance	$R_{DS(on)1}$	$I_D=-25A, V_{GS}=-10V$		13	17	$m\Omega$
	$R_{DS(on)2}$	$I_D=-13A, V_{GS}=-4.5V$		18.5	26	$m\Omega$
Input Capacitance	C_{iss}	$V_{DS}=-20V, f=1MHz$		2400		pF
Output Capacitance	C_{oss}			330		pF
Reverse Transfer Capacitance	C_{rss}			240		pF
Turn-ON Delay Time	$t_{d(on)}$	See specified Test Circuit.		17		ns
Rise Time	t_r			260		ns
Turn-OFF Delay Time	$t_{d(off)}$			190		ns
Fall Time	t_f			160		ns
Total Gate Charge	Q_g	$V_{DS}=-20V, V_{GS}=-10V, I_D=-50A$		47		nC
Gate-to-Source Charge	Q_{gs}			11.5		nC
Gate-to-Drain "Miller" Charge	Q_{gd}			8		nC
Diode Forward Voltage	V_{SD}	$I_S=-50A, V_{GS}=0V$		-1.02	-1.5	V

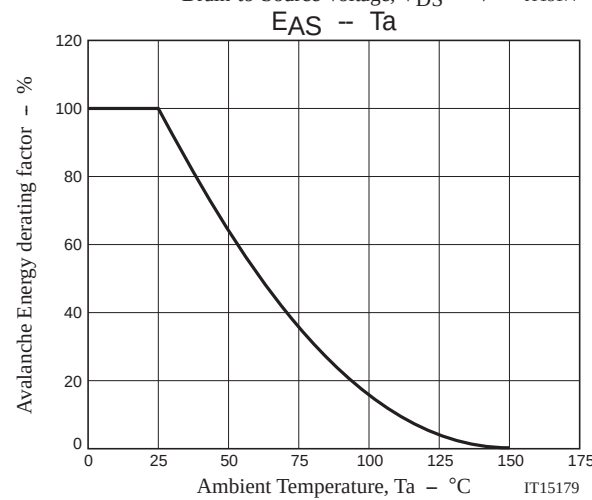
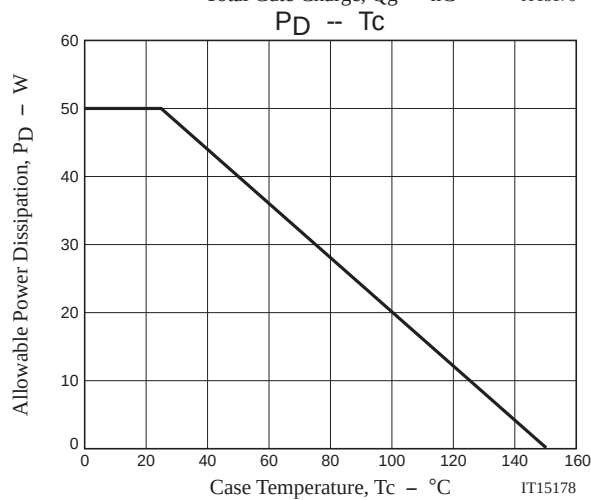
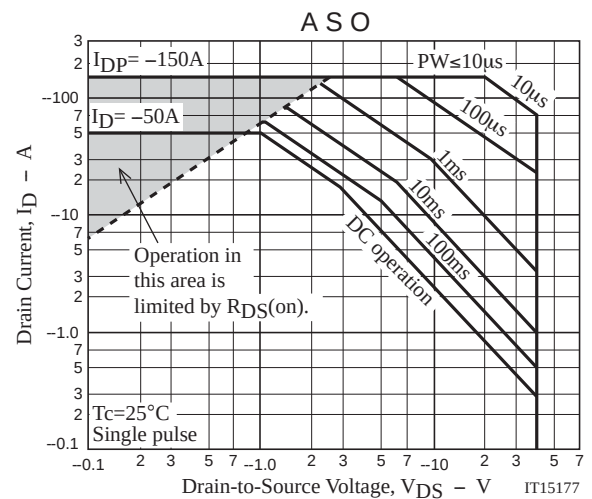
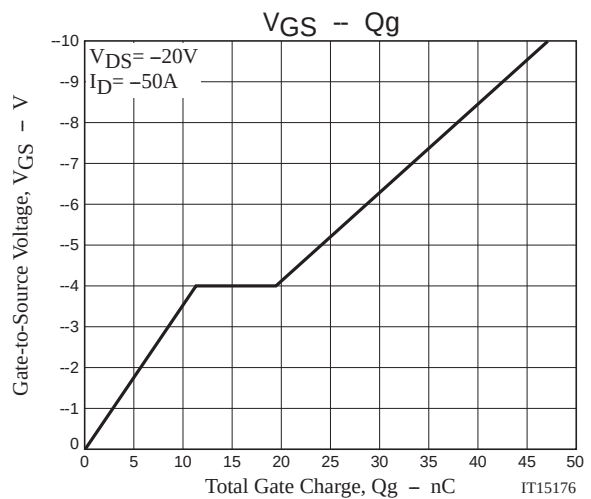
Switching Time Test Circuit



Ordering Information

Device	Package	Shipping	memo
ATP107-TL-H	ATPAK	3,000pcs./reel	Pb Free and Halogen Free





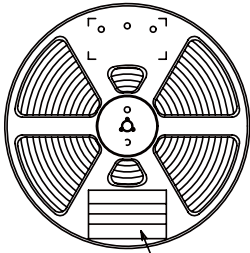
Taping Specification

ATP107-TL-H

1. Packing Format (TL)

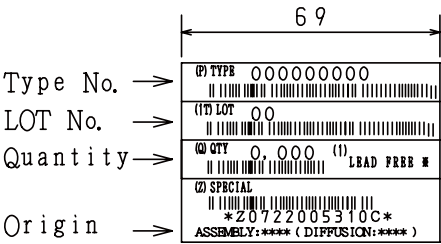
Package Name	Carrier Tape Type	Maximum Number of devices contained (pcs)			Packing format	
		Reel	Inner box	Outer box	INNER BOX SD-C-18	OUTER BOX SD-A-18
ATPAK	ATP	3,000	3,000	15,000	1 reels contained Dimensions:mm (external) 340×340×28	5 inner boxes contained Dimensions:mm (external) 355×355×165

Packing method



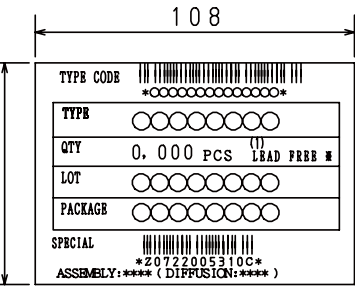
Reel label

Reel label, Inner box label
(unit:mm)



Outer box label

It is a label at the time of factory shipments.
The form of a label may change in physical distribution process.



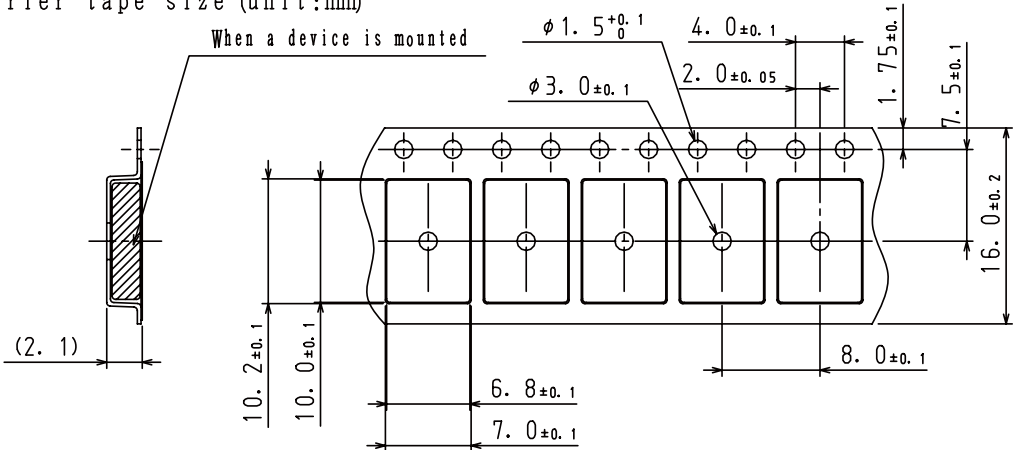
NOTE (1)

The LEAD FREE * description shows that the surface treatment of the terminal is lead free.

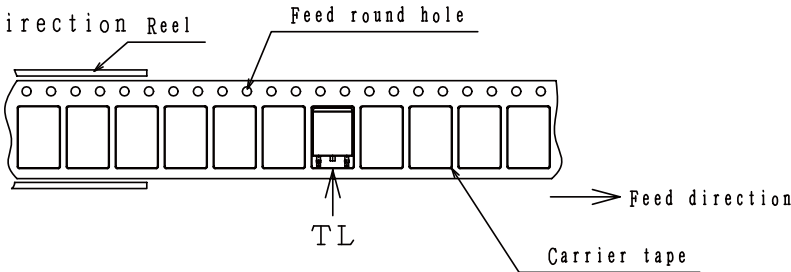
Label	JEITA Phase
LEAD FREE 3	JEITA Phase 3A
LEAD FREE 4	JEITA Phase 3

2. Taping configuration

2-1. Carrier tape size (unit:mm)

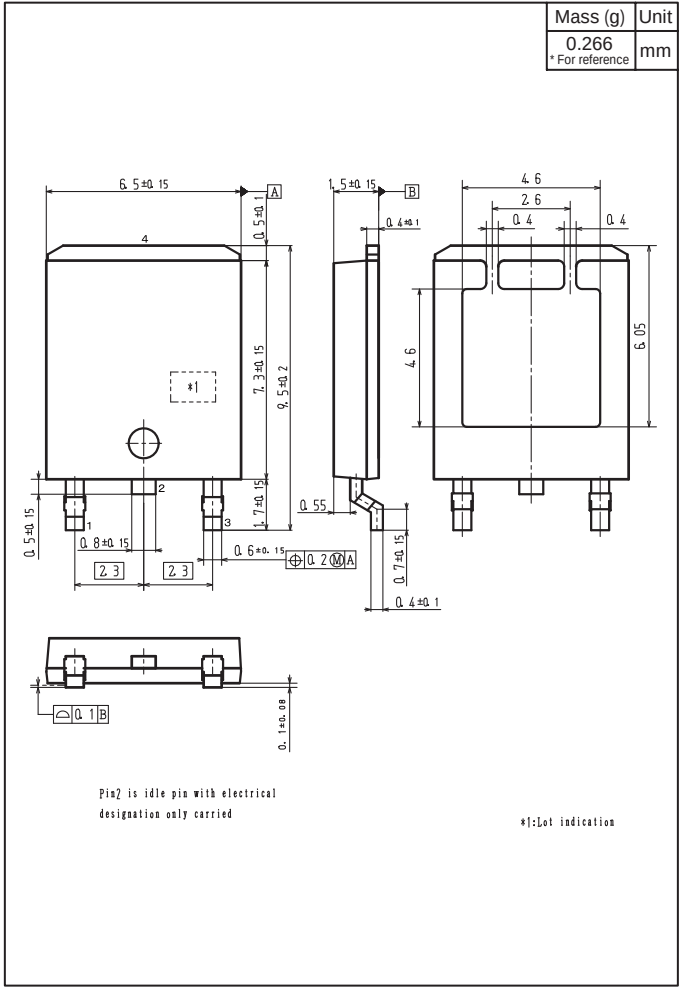


2-2. Device placement direction Reel

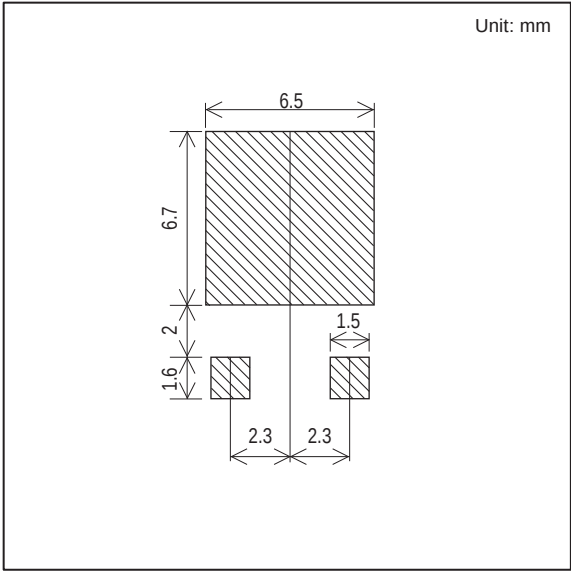


The one electrode terminals on feed hole side...TL

Outline Drawing
ATP107-TL-H



Land Pattern Example



Note on usage : Since the ATP107 is a MOSFET product, please avoid using this device in the vicinity of highly charged objects.

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